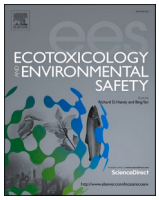
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Review



Analysis of the long-term effectiveness of biochar immobilization remediation on heavy metal contaminated soil and the potential environmental factors weakening the remediation effect: A review

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ABSTRACT

Currently, the research and application of biochar in the remediation of heavy metal contaminated soil has become a hotspot, especially regarding the remediation of agricultural land. Biochar has been proved to be effective in reducing the content of available heavy metals in the soil as well as the heavy metals in plants. However, the long-term effectiveness of biochar immobilization has not been widely studied. In this review, retrospective search was carried out on the published literature results concerning remediation effects of biochar on different areas of heavy metal contaminated soil in the recent years, its application in field remediation (several years), and some potential abiotic and biotic factors that may weaken the immobilization effects of biochar. This results indicate that: (1) biochar is widely used in the remediation of heavy metal contaminated soil in different areas and has excellent immobilization effect. (2) Most of the research demonstrate that the immobilization effect of biochar is effective for 2–3 years or according to few results even for 5 years. However, there have been various reports claiming that the immobilization effect of biochar decreases with time. (3) Abiotic factors such as acid rain, flooded environment, changes in soil condition (pH, redox and dissolved organic matter) and changes in biochar (Cl^- and alkali leaching) can significantly weaken the immobilization effect of biochar. (4) Biotic factors such as plant roots, earthworms and soil microorganisms can also significantly reduce the immobilization effect of biochar. Therefore, field experiments having longer time span with biochar need to be further carried out, and the developmental research of modified biochar with a more stable immobilization effect also needs further attention.

1. Introduction

Biochar is a solid material with high aromaticity and high carbon content (more than 60% carbon content) which is obtained by thermochemical conversion of biomass under the presence of limited oxygen (Initiative et al., 2012). In the recent years, biochar has become one of the most attractive research hotspot as a result of its special properties along with its important role in the climate change, global biogeochemical cycle and environmental system (Lehmann et al., 2006). Biochar plays important roles in improving the property of the soil such as soil quality improvement and nutrient retention, crop yield improvement (Ding et al., 2016; Agegnehu et al., 2017), as well as increment of the soil carbon sink and reduction of the carbon dioxide emissions (Li et al., 2018). Moreover, it also finds wide applications for the remediation of soil pollution including remediation of heavy metals and organic pollutants (O'connor et al., 2018; Dai et al., 2019). Compared to the conventional soil remediation methods, biochar not only plays an important role in immobilization remediation of organic soil pollutants (antibiotics, polycyclic aromatic hydrocarbons (PAHs), polychlorinated biphenyls (PCBs), etc.) and heavy metal pollutants (Cd, Pb, Zn, etc.) but

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as an economical product, also provides a good treatment scheme for organic waste matter from industrial and agricultural processes (Ahmad et al., 2014). A large variety of studies, including pot and field experiments, have reported the excellent effect of biochar on the remediation of heavy metal contaminated soil, whether in farmland soil, mining area soil or urban soil (He et al., 2019a,b; Karami et al., 2011; Netherway et al., 2019; O'connor et al., 2018). For example, Zhang et al. (2016) observed significant reduction of Cd content in the rice grains by applying sludge biochar in the field. Nie et al. (2018)was able to significantly reduce content of Cd (62%–76%), Pb (17.3%–49.1%) and Cu (15%–38%) in *Brassica* using bagasse derived biochar. However, there are very few studies focusing on the long-term effectiveness of biochar immobilization remediation in the field experiments with most of them focusing for two to three years. Such as Bian et al. (2014) conducted farmland immobilization experiment with biochar in Yixing, Jiangsu Province for a period of three years from 2010 to 2012. Sui et al. (2018) carried out a field immobilization experiment in Jiyuan, Henan province for a period of three-years.

However, immobilization remediation materials, such as biochar, only change the existing form of the heavy metals in soil without reducing the total amount of heavy metals in the soil (O'connor et al., 2018). At present, the evaluation of the immobilization remediation effect of biochar is more inclined to consider the short-term effects using the accumulation of heavy metals during the growth period of the crops or the change of the heavy metals form in the soil as indicators. At present, there is dearth of studies about the long-term effectiveness of biochar in the field application. Moreover, researchers have also demonstrated that a part of heavy metals adsorpted by biochar could be re-released (Uchimiya et al., 2011; Melo et al., 2016), therefore, heavy metals adsorbed and passivated by biochar are at risk of being re-released under the influence of some environmental factors.

Furthermore, the biochemical stability of pyrolytic carbon, such as biochar, is being questioned with growing number of evidences suggesting the stability to be much lower than previously anticipated (Knicker et al., 2013; Watzinger et al., 2014). Many studies have shown that the biotic or abiotic processes in the soil can lead to degradation of some biochar (Cheng et al., 2008; Guggenberger et al., 2008; Spokas, 2010), with only the stable carbon components being preserved over the long term (Zimmermann et al., 2012). In addition, the process of biotic or abiotic degradation will significantly change the physical and chemical properties of the biochar (de la Rosa et al., 2018). Furthermore, the soil in the field is also vulnerable to the strong intervention of various climate and human activities, such as acid rain (Wang et al., 2018), flooding (El-Naggar et al., 2019), tillage (Chang et al., 2014), planting (Oustriere et al., 2017), and so on. These effects from climate change and human activities lead to changes in the soil conditions which may significantly affect the form distribution of heavy metals and consequently the long-term effectiveness of biochar immobilization may also be affected. In addition, some previous studies have also indicated that the soil organisms significantly affect the heavy metal forms in the soil by changing the physical and chemical properties of soil (Chaudhary et al., 2019; Wen et al., 2004; Wang et al., 2019a,b,c). For example, earthworms weaken the immobilization remediation effect of biochar by reducing the pH and increasing the available heavy metal content in the soil (Wang et al., 2019a,b,c). It has also been reported that in the plant rhizosphere environment the immobilization effect of biochar has been significantly weakened (Houben and Sonnet, 2015), and that the root exudates like small molecule organic acids also cause re-release of the adsorbed heavy metals from the biochar (Alozie et al., 2018).

Therefore, the present review aims to summarize the immobilization remediation effect and long-term effectiveness of biochar on heavy metal contaminated soil (more than two years of experiments). Furthermore, we have also tried to discuss some potential biotic or abiotic factors of the environment which may influence the biochar remediation as per the existing reports. Our manuscript also provides some theoretical references for the future application and research of biochar as immobilization remediation agent for soil pollution and puts forward some reasonable and feasible research directions.

2. Effect of preparation of biochar to its immobilization

Biochar is the carbonized product of organic matter and there are various kinds of raw materials available for the preparation of biochar. From the theoretical perspective all organic substances can be used as raw materials for the preparation of biochar. Current research and production are mainly based on agricultural waste [e.g. rice straw, wheat straw, corn stalk, cotton stalk and grass](Jeong et al., 2016; Kloss et al., 2012; Luo et al., 2015; Ren et al., 2015), forest waste [e.g. pine waste, and palm waste etc.,](Mukherjee et al., 2011; Uchimiya et al., 2013), livestock waste [e.g. pig manure, and cow manure, etc.,](Chen et al., 2020) and municipal sewage sludge (Lu et al., 2013). The biochars, obtained from different raw materials, exhibit widely different physical and chemical properties, with the properties being dependent also on other factors like pyrolysis mode, pyrolysis temperature and pyrolysis time (Wang et al., 2020). Currently, the thermochemical processes include pyrolysis (fast and slow), gasification, torrefaction and microwave pyrolysis (Ok et al., 2018). In the recent years, hydrothermal carbonization of biomass has also been studied in China and outside world (Wang et al., 2019a,b,c; Zhang et al., 2019). Sangani et al. (2020) found that three different raw materials, each having four different particle sizes, can produce 12 kinds of biochars with different properties, by pyrolysis. In addition, various studies have reported the important effect of production temperature on different properties of the produced biochars including the pH, cation exchange capacity, specific surface area, surface functional groups and mineral concentration (Sizmur et al., 2017; Zhang et al., 2018). With the increase of pyrolysis temperature, it has been observed that the absorption peaks corresponding to –OH, C–O–C, $–CH_2-$, C═O present on the surface of biochar decrease gradually. Only the peaks corresponding to C═C and C═O present on the aromatic ring were found to be well preserved during high temperature biochar preparationChen and Chen, 2009.

At present, the immobilization effect of biochar on the heavy metals of soil is reported mainly through two methods (He et al., 2019a,b): (1) Direct adsorption by biochar via electrostatic attraction, complexation, ion exchange, precipitation and physical adsorption; (2) Improving the adsorption capacity of the soil particles by changing the physicochemical properties (soil pH, CEC, mineral and organic matter content), followed by adsorption via biochar as shown in Fig. 1. Therefore, it can be said that the preparation conditions of biochar determine the physical and chemical properties of biochar and also indirectly determine adsorption and immobilization effect of biochar to the heavy metals (Fig. 1).

3. The remediation effect of biochar for heavy metal contaminated soil

Soil heavy metal pollution is widespread in most areas of China. According to the survey bulletin by the Ministry of Environmental Protection of China in 2014, the over-standard rate of overall soil in China reached 16.1%. (MEP, 2014). At present, as an environmental friendly carbon material, biochar is a research hotspot in the remediation of heavy metal contaminated soil. However, the application of biochar in other heavy metal polluted site is not as common as in farmland. The soil properties and heavy metal species of different pollution sites are greatly different, which may affect the remediation effect of biochar. For this reason, we reviewed the remediation effect of biochar on different heavy metal pollution sites below (Table 1). In addition, we reviewed the long-term remediation experiments of biochar and discussed the long-term effectiveness of biochar remediation.

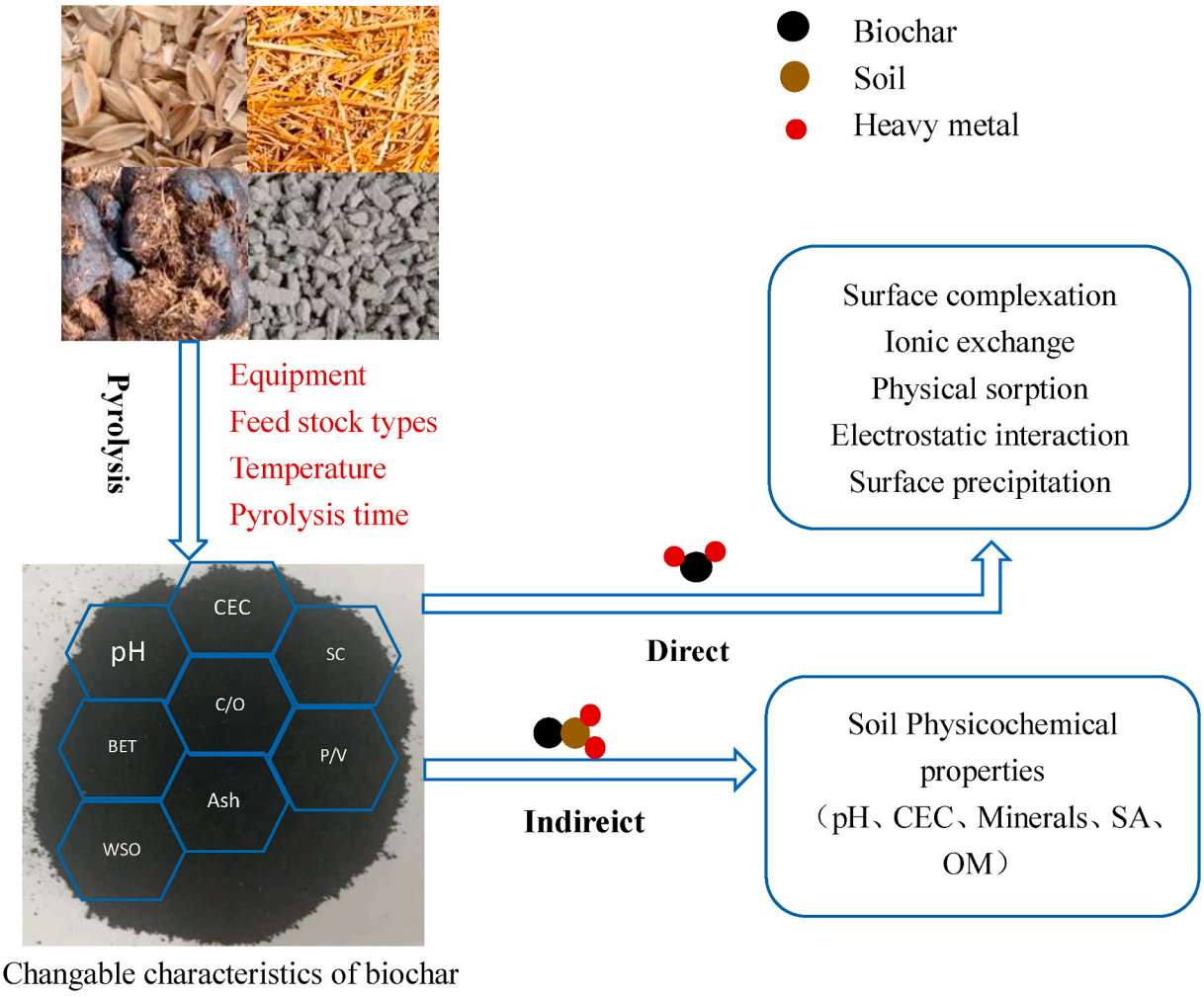


Fig. 1. The schematic diagram of the effect of the preparation conditions on the immobilization effect of biochar. WOS: water soluble organic; WSI: water soluble ions; SC: surface charge; P/V: pore/volume; SA:Surface area; OM: Organic matter (Mandu et al., 2015; He et al., 2019a,b; Wang et al., 2020).

3.1. Study on the remediation of biochar for heavy metal contaminated soil in different areas

The heavy metal contaminated sites mainly include mining areas, farmland areas and urban industrial areas (MEP, 2014). Currently, wide application of biochar is being observed for the remediation of heavy metal contaminated soil in different site (Table 1). For example, Karami et al. (2011) studied a combination of biochar and compost to restore the contaminated soil from a copper mine in Cheshire, England. The results showed that biochar had a good immobilization effect on both Cu and Pb. Netherway et al. (2019) studied the remediation effect of phosphorus-rich biochar on heavy metal contaminated soil in the cities, and found significant reduction in the content of availabe Pb in the soil by the phosphorus-rich biochar. Gan et al. (2012) explored the remediation effect of straw biochar on the contaminated soil of electroplating plant. The results showed that straw biochar had an obvious stabilizing effect on the heavy metals. Additionally, heavy metal pollution in arable land has a more direct impact on the food security and human health. Therefore, the prospect of remediation of heavy metal polluted farmland via the application of biochar is another hot research topic with respect to the remediation of contaminated soil. For example, Zhan et al. (2019) applied biochar to farmland around a Pb–Zn mining area (Yunnan Province, China), and found out significant increase in the contents of soil nutrient elements; there was significant reduction in the contents of available Pb and Cd in the soil and contents of Pb and Cd in the plants. However, it has also been reported that the application of biochar significantly reduced the accumulation of Cd, Pb and Zn by rice, but simultaneously increased the accumulation of As (Zheng et al., 2012). There are similar results in Wang et al. (2017), the application of biochar promoted the release of As in paddy soil. This risks needs to arouse people's attention in the application of biochar. In addition, during the general experimental process, the concentration of biochar is unrealistically high (1%–20%) and in this regard Zhang et al. (2016) tried to use low concentration of biochar in their experiments. They found out that sludge biochar, in the concentration range of 1.5 t/ha and 3 t/ha, could significantly reduce the Cd content in the rice grains grown in Cd contaminated farmland and increase the biomass content of rice. However, the remediation effect of other biochars at low doses remains to be studied. In general, we believe that biochar is a amendment that can be widely used to remediate various contaminated soil sites by heavy metals.

3.2. Study on the remediation of biochar in multiple years

At present, a large part of biochar research is still conducted on the indoor potted plants with field experiments for short-time duration, though small numbers of researchers have also carried out field restoration experiments of biochar for several seasons or years. For this purpose, this article included the summary of research on biochar remediation conducted over long duration along with the statistics (Table 2). It was found that the biochar remediation experiments were mainly carried out for 2–3 years with the rice being the major crop. Most

Table 1
Literature results concerning remediation effects of biochar on different areas of heavy metal contaminated soil.

	Soil source	Feedstock/temperature	Application rate	Soil remediation effect	References
Mine soil	A zinc mining area located in Vazante, Minas Gerais state, Brazil	Wood biochar from Eucalyptus sp.	0, 2.5 g/kg, 7.5 g/kg, 15 g/kg and 30 g/kg	Increased soil pH; reduced plant extractable heavy metals; promoted seed germination.	Martins et al. (2018)
	Located in an old copper mine in Touro, north-west Spain	Quercus ilex wood 400°C	0 and 4%	Reduced the phytoavailable metal concentrations.	Forjan et al. (2018)
	Coal mine, Makum coalfield, Margherita, Assam situated in the North-Eastern part of India	Distilled waste of lemongrass (Cymbopogon flexuosus) 450°C	0, 1%, 2% and 4%	Reduced soil acidity and bioavailability of metal; improved plant productivity and soil health.	Jain et al. (2019)
	Zinc mining area in Vazante (Minas Gerais	Sugarcane straw 700°C	0, 1.5%, 3.0% and 5.0%	Decreased the available Cd, Pb, and Zn concentrations in the soil; reduced accumulation of Cd, Pb and Zn in the plant.	Puga et al. (2015)
	A Zn mining area located in Vazante, State of Minas Gerais (MG), Brazil	Sewage sludge 500°C Wood (*Eucalyptus* sp.) 350°C	WB: 30 g/kg and 60 g/kg; SS: 10 g/kg and 20 g/kg	Increased leachate and soil pH; reduced the concentration of bioavailable Cd, Pb and Zn concentraion.	Penido et al. (2019)
	The Cu-mining area of Riotinto, Spain	Manure waste 450 and 600°C	0 and 10%	Increased the biomass of *Brassica napus*; reduced the accumulation of heavy metals by *Brassica napus*.	Gasco et al. (2019)
	The Formosa mine site is located 16 kms south of the town of Riddle, OR, USA	Miscanthus 700°C	0, 1%, 2% and 5%	Increased soil pH; reduced DOC content and heavy metal content in filtrate.	Novak et al. (2018)
	Mining site at Cave del Predil (NE, Italy)	Prune residues; fir tree pellets; manure pellets (70%) mixed with fir tree pellets (30%) 500°C	0, 1.5% and 3%	Reduced plant accumulation of Cd and Pb; increased biomass of *Anthyllis vulneraria* and *Poa alpina*; reduced the concentration of bioavailable heavy metals.	Fellet et al. (2014)
	Raibl mine in the Rio del Lago valley (northern side of Julian Alps, NE, Italy)	Prune residues from orchards	0, 1%, 5% and 10%	Increased the pH, the nutrient retention, CEC and the water holding capacity; decreased the bioavailability of Cd, Pb, Ti and Zn in the mine tailings.	Fellet et al. (2011)
	An E-waste recycling site located in Qingyuan city, Guangdong province, China	Dairy manure 300°C	0, 2.5% and 5%	Increased the soil pH, CEC and available phosphorous; significantly reduced the bioavailable Cd,Pb,Cu and Zn concentration.	Chen et al. (2018)
Urban soil	An anonymous contaminated site in urban Victoria, Australia.	Poultry litter and Biosolids 300, 400 and 500°C	0, 1% and 3%	Significantly reduced the concentration of bioavailable Pb; biochars were able to outperform phosphate amendments for Pb immobilization.	Netherway et al. (2019)
	A military firing range was collected from Busan Metropolitan City in the Republic of Korea	Soybean stover 700°C	1–20%	Significantly reduced the concentration of TCLP-Pb in soil and highest reached 90%.	Moon et al. (2013)
	A site located in Castleford, West Yorkshire, UK(various chemical industries)	British broadleaf hardwood 600°C	0%, 0.5%, 1% and 2%	Significantly reduced the concentration of the extracted concentrations of Ni and Zn by 83%–98% over three years	Shen et al. (2016)
	A military shooting range form Cheorwongun in Gangwon-do Korea	Soybean stover and pine needles 300 and 700°C	0 and 10%	Both biochars significantly reduced the concentration of bioavailable Cu and Pb, and the effect of soybean straw biochar was better.	Ahmad et al. (2016)
	An Electroplating Factory in China Southern Jiangsu Province	wheat straw	0, 1 g/kg, 3 g/kg, 5 g/kg, 7 g/kg and 10 g/kg	Biochar changed the forms of heavy metals in the soil and had significant effect immobilizing the polluted soil.	Gan et al. (2012)
	A cultivated upland near an abandoned mining area	Rice hull 500°C	0, 0.5%, 1%, 2%, 5% and 10%	Increased soil pH; Significantly reduced NH_4NO_3-heavy metal in soil; Reduced heavy metal concentration in lettuce.	Kim et al. (2015)
	A private farm at MetKenana village, Touch district, Qalyubia Governorate, Egypt.	Sugarcane bagasse 400–450°C	0, 10 t/ha, 15 t/ha, 20 t/ha and 30 t/ha	Reduced the soluble/exchangeable fraction of Cd; increased biomass of soybean.	Mohamed et al. (2019)
Farmland soil	A rice farm located in Feilongqiao village, Hunan province in southern China	Sewage sludge 500°C	0, 1.5 t/ha and 3 t/ha	The Cd concentration in rice grains was significantly reduced by 39.25%–40.74%	Zhang et al. (2016)
	A paddy field adjacent to a mine in Youxi County, Fujian Province, China.	Rice straw 500°C	0, 2.5% and 5%	Reduced $CaCl_2$-extractable Cd concentration in soil.	Li et al. (2019)
	A rice paddy in Zhuzhou, Hunan province	Water hyacinth and rice straw 450°C	0, 2% and 5%	Reduced $CaCl_2$–Cd and Pb concentration in soils with the reduction being increased with elevated BCs rate.	Yin et al. (2016)
	A rice field in Xinma Town, Zhuzhou, Hunan Province, China	Bean stalk and rice straw 500°C	0 and 20 t/ha	Decreased NH_4NO_3–Cd, Zn and Pb concentrations in soil by 66.0%–90.1%, 73.1%–92.2%, and 74.1%–91.0%, respectively; reduced the concentration of heavy metals in rice.	Zheng et al. (2013)
	Farmer fields near a mining area in Zhuzhou city, Hunan province, south China	Rice straw, husk and bran 500°C	0 and 5%	Decreased concentrations of Cd, Pb and Zn by up to 98%, 72% and 83% in rice shoot; decreased concentrations of Cd and Zn in the pore water.	Zheng et al. (2012)

Table 2
Study on the remediation of heavy metal contaminated farmland by biochar (multiple years).

Feedstock/ temperature	Location	Application rate	Planting	Time	Contamination	Soil remediation effect	References
Wheat straw 350–550°C	Yixing Jiangsu (China)	0, 10 t/ha, 20 t/ha and 40 t/ha	Wheat	2009–2010 two years	Cd: 21.84 mg/kg	Decreased $CaCl_2$–Cd by 10.1%–40.2% in 2009, 10.0-57.0% in 2010; reduced rice grain Cd concentration by 24.8%–44.2% in 2009, 14.0%–39.2% in 2010.	Cui et al. (2012)
Wheat straw 350–550°C	Yixing Jiangsu (China)	0, 10 t/ha, 20 t/ha and 40 t/ha	Rice	2009–2010 two years	Cd: 21.84 mg/kg	Decreased $CaCl_2$–Cd by 32.0%–52.5% in 2009 and 5.5%–43.4% in 2010; decreased rice grain Cd concentration by 16.8%, 37.1%, 45.0%, respectively, in 2009 and 42.7%, 39.9%, 61.9%, respectively, in 2010.	Cui et al. (2011)
Palm(iron-modified) 300°C	Qingyuan, Guangdong (China)	0 and 1.5 t/ha/ each season	Rice	2013–2014 Two years (four season)	Cd: 2.86 mk/kg As: 22.6 mg/kg	Increased the soil pH; decreased available As and available Cd in the soil; reduced the accumulation of As and Cd in rice grains.	Pan et al. (2019)
Wheat straw 350 and 550°C	Yixing Jiangsu (China)	0, 10 t/ha, 20 t/ha and 40 t/ha	Rice	2010–2012 Three years	Cd: 5mg/kg Pb: 100mg/kg	Increased pH, organic matter of soil and immobilized soil Cd and Pb; consistently reduced rice Cd and Pb content in three years.	Bian et al. (2014)
Miscanthus 860°C	Berlin (Germany)	0, 1%, 2.5% and 5%	orchard grass (*Dactylis glomerata* L.)	2011–2012 two years	Cd: 6.8–99 mg/kg Pb:153–249 mg/kg Zn:368–780 mg/kg	Increased biomass production; reduced Zn and Cd concentrations and increased Cu and Pb concentrations in the soil solution;	Wagner et al. (2015)
Wheat straw 485°C	Jiyuan Henan (China)	0, 20 t/ha and 40 t/ha	Wheat-corn rotation	2013–2016 Three years	Cd: 1.78 and 3.81 mg/kg; Pb: 118 and 230 mg/kg	Increased grain yield; reducded Cd and Pb concentrations in wheat grain in 2014; but, biochar may have reduced efficacy for reducing uptake of heavy metal in wheat in 2015–2016.	Sui et al. (2018)
Corn stalks, peanut husks and rice husks 450°C	Shenyang, Liaoning (China)(Potted)	0, 1%, 2% and 4%	Rice	2012–2014 Three years	Cd: 3.98 mg/kg	Reduced the exchangeable Cd concentration by 28.5%–59.4% in soil and the total Cd accumulation in rice grain by 2.7-23.8%.	He et al. (2016)
Poplar woodchips 550°C	Arnoldstein, Carinthia (Austria)	0, 1% and 3%	Miscanthus × giganteus rhizomes	2013–2014 Two years	Cd: 6.65-11.0 mg/kg Pb: 1124–1148 mg/kg Zn: 1151–1852 mg/kg	Reduced NH_4NO_3–Cd, Pb and Zn concentrations by 75%, 86% and 92%.	Karer et al. (2018)
Holm oak 650°C	Campine (Belgian)	0, 2% and 4%		2015–2017 Three years	Cd: 11 mg/kg Pb: 247 mg/kg Zn: 298 mg/kg	Decreased concentration of Cd and Zn by 40%–66% and 48%–77%, respectively, in pore water.	Egene et al. (2018)
Pine bark 420°C	Gironde, (France)	0 and 1%	P. nigra and Giant reed	2014–2016 Two years	Cu: 964 mg/kg	Reduced the concentration of Cu^{2+} in soil pore water.	Oustriere et al. (2017)
Rice straw and coconut husk	Shaoguan, Guangdong (China)	0, 2.25 t/ha, 6.75 t/ha, 11.25 t/ha and 22.5 t/ha	Rice	2014–2015 Two years (four season)	Cd: 0.42 mg/kg As: 35.4 mg/kg	Improved the rice yields; increased soil pH and total organic carbon; decreased the Cd content in rice grains over the four growth seasons.	Chen et al. (2019)
Salisbury biochar 600°C	West Yorkshire, (UK)	0, 0.5%, 1% and 2%		2011–2014 Three years	Ni:200–740 mg/kg Zn:150–810 mg/kg	Reduced the extracted concentrations of Ni and Zn by 83%–98% over three years	Shen et al. (2016)
Wheat straw 485°C	Xiangyin, Henan (China)	0, 20 t/ha and 40 t/ha	Rice (high and low accumulation)	2013–2014 Two years (four season)	Cd: 0.9 mg/kg Zn: 171 mg/kg	No significant effect the first season; Consistently reduced rice grain Cd by a maximum of 61%, 86% and 57% over the next three seasons.	Chen et al. (2016)
Corn stalks and hardwood 300°C	Tongling, Anhui (China)	0, 3% and 5%	Rice	2012–2014 Three years	Cd: 2.2 mg/kg Cu: 452 mg/kg	Both biochars can reduced the $CaCl_2$–Cd and Cu in soil, and Cd and Cu uptake in rice tissue respectively; but the stability of the two biochar effects is greatly different.	Li et al. (2016)
Rape straw and rice straw 300°Cand 550°C	Linxiang, Hunan (China) (Potted)	0, 2% and 5%	Cabbage and wheat	2015–2017	Pb: 742.40 mg/kg Cu: 149.68 mg/kg	Reduced the phytoavailable Pb and Cu in soil; decreased concentration of Pb and Cu in plants by 40.81% and 56.14% respectively.	Salam et al. (2019)
Kenaf and Sludge 350 and 550°C	Guangdong (China)	600 g soil mixed with 24 g biochar		Two years	Pb: 2050 mg/kg Zn: 495 mg/kg Cu: 654 mg/kg Cd: 0.705 mg/kg As: 1177 mg/kg	Effectively immobilized Pb and Cu but not Zn, Cd, and As; the immobilization capacity of biochar changed during the long-term aging.	He et al. (2019a,b)
Municipal Biowaste 450–550°C	Yixing, Jiangsu (China)	0 and 40 t/ha	Rice-wheat rotation	2010–2011 Two years	Cd: 0.47 mg/kg Pb: 53.8 mg/kg Cu: 31.1 mg/kg Zn: 58.3 mg/kg Ni: 34.6 mg/kg	Decreased the soil availabilty of Cd but not influenced the soil availability of Pb, Cu, and Ni; changes in heavy metal concentration in rice and wheat grains have a similar trend.	Bian et al. (2014)
			Rice		Cd: 0.85 mg/kg		

(*continued on next page*)

Table 2 (*continued*)

Feedstock/ temperature	Location	Application rate	Planting	Time	Contamination	Soil remediation effect	References
Rice straw 450°C	Xiangyin, Hunan (China)	0, 10 t/ha and 20 t/ha		2016–2017 Two years (three seasons)		Decreased rice grain Cd concentration by 30.4%–47.9%, 45.0%–50.2% and 8.6%-32.1 %, respectively, in three rice seasons; reduced the $CaCl_2$–Cd concentration in the soil over three seasons.	Sui et al. (2020)
Wheat straw 450°C	Yixing Jiangsu (China)	0, 10 t/ha, 20 t/ha and 40 t/ha	Rice-wheat rotation	2009–2013 Five years	Cd: 22.65 mg/kg Pb: 621.23 mg/kg	Decreased bioavailable Cd and Pb in a contaminated paddy field during five years	Cui et al. (2016c)

data showed that the biochar application reduced the bioavailability of heavy metals, and its immobilization effect has long-term effectiveness (Table 2). For example, Cui et al. (2011, 2012) conducted a biochar field experiment on Cd contaminated soil for two-years. The biochar contents of 10 t/ha, 20 t/ha and 40 t/ha were found to reduce the Cd content in rice and wheat grains by 16.8%–45.0% and 24.8%–44.2%, respectively, in the first year, whereas the corresponding reductions in the second year were 39.9%–61.9% and 14.0%–39.2%, respectively. Bian et al. (2014) reported that biochar could effectively reduce the content of bioavailable Pb and Cd in the polluted farmland in three years by carrying out a three-year biochar remediation experiment. Cui et al. (2016a) found that the bioavailability of Cd and Pb was significantly reduced by 8%–45% and 14%–50%, respectively, during the five years of experimental period. Surface adsorption, oxygen-containing functional groups adsorption, mineral phases (Al, Fe and P) precipitation and cation exchange play a major role in the immobilization (Cui et al., 2016a, 2016c).

However, in the study of Chen et al. (2016b), no significant reduction in the Cd content of the rice grains by wheat straw biochar was observed in the first season though there was significant reduction in the content of $CaCl_2$–Cd and Zn in the soil at the beginning. However, in the following three seasons, the Cd content in the rice grains was reduced by 61%, 86% and 57%, respectively. The authors concluded that the availability of biochar may also be related to the rice varieties and the yearly environment of the paddy field. In addition, biochar in the concentration range of 20 t/ha in the first, second and fourth seasons significantly reduced soil $CaCl_2$–Cd content by 48%, 60% and 20%, respectively. The biochar dosage of 40 t/ha significantly reduced the $CaCl_2$–Cd content by 70%, 85%, 54% and 43% in four successive seasons, indicating initial increase of the immobilization effect of biochar during the field aging process followed by a decrease (Chen et al., 2016a). Li et al. (2016) reported similar results from their field experiments (three years). The immobilization effect of the hardwood biochar was reduced in the second and third years while that of the corn biochar was stable, suggesting that long-term stability of the biochar immobilization effect is closely related to the biochar type. Another explanation given by some researchers for the reduction of biochar immobilization effect in the second and third years during the field experiment of three years is the change of climatic condition; the wheat growing season was very rainy in 2015–2016, while in 2014 it was relatively dry (Sui et al., 2018). In the meanwhile, some studies reported along with the reduction of the content of available heavy metals in the soil, there was significant increase in the content of heavy metals in the above-ground part of the crops due to application of the biochar (Zhang et al., 2017; Li et al., 2019a, 2019b). Therefore, the long-term effectiveness of immobilization remediation due to biochar in the field may be significantly affected by some environmental factors. Therefore, we analyzed some potential environmental factors that may lead to the weakening of the immobilization adsorption effect of biochar.

4. Potential factors affecting the long-term effectiveness of biochar

It is well known that the immobilization effect of biochar is determined by the physicochemical properties of biochar along with the environmental conditions of the soil. The physical and chemical properties of biochar under natural conditions are changed by various environmental factors (Rechberger et al., 2017). For example, high pH of the biochar is one of the important factors responsible for biochar immobilization. However, slow surface oxidation and leaching of alkaline minerals will reduce the alkalinity of the biochar, thereby reducing the immobilization performance of the biochars towards heavy metals (colloids) (Cui et al., 2016b). Additionally, in the field conditions, various environmental factors including biotic and abiotic factors, constantly affect the soil after biochar improvement. The change of soil environment can also significantly change the transfer of heavy metals. Therefore, the long-term stability of biochar due to environmental factors and its potential impact on the immobilization effect is a prerequisite in order to discuss the long-term effectiveness of immobilization remediation by biochar. In this review, the long-term stability of biochar in the field environment, the potential biotic and abiotic factors in the environment that may weaken the immobilization effect of biochar are discussed.

4.1. Long-term stability of biochar

The stability of biochar refers to its ability to resist biotic and abiotic degradation (chemical oxidation, photochemical oxidation, inorganic pyrolysis, etc.) in the soil environment, which is one of the preconditions for use of biochar as a means for reduction of greenhouse gas emissions and slowing down climate warming (Wang et al., 2016). On one hand, high degree of carbonization of biochar and compact aromatic structure makes this structure chemically stable. On the other hand, there are abundant organic structures on the surface of biochar which can form organic-inorganic complexes with the minerals in the soil (soil agglomeration), so as to reduce the risk of biochar decomposition by soil microorganisms and thereby achieving a stable structure (Pan et al., 2007). Usually, biochar has long been considered as a material with high stability. The natural degradation of biochar itself will be very slow, with an expected half-life of 102–107 years for biochar C in the environment (Zimmerman, 2010). Thus, biochar can be considered inert within the time frame of most soil remediation projects, and thus has the long-term effectiveness of immobilization for heavy metal in soil (Sizmur et al., 2016).

Unfortunately, more and more evidences suggest that the biochemical stability of pyrolytic carbon, such as biochar, is much lower than previously thought (Knicker et al., 2013; Watzinger et al., 2014). After the biotic and abiotic degradation in the soil, only the stable carbon components can be preserved for a long time (Cheng et al., 2008; Guggenberger et al., 2008; Spokas, 2010; Zimmermann et al., 2012). Through ^{13}C and ^{15}N nuclear magnetic resonance (NMR) spectra, de la Rosa and Knicker (2011) found significant change in the chemical structure of the peat after hatching for two months, which included

degradation of the N-heterocyclic components and oxidation of the aryl structures. It also shows that even within a short time period, pyrolytic carbon has obvious degradation. Later, de la Rosa et al. (2018) further found that in the presence of wild environment the particle structure was largely broken within 24 months. A large amount of aromatic C was converted into O-alkyl or C-alkyl groups, with C loss reaching up to 27%. Thus, the stability of biochar was far lower than expected. On the other side, Quan et al. (2020) pointed out that the microbial aging reduced the stability of biochar, with the fulvic acid and humic acid analogs being the main components of the soluble part of biochar after aging. Overall, due to the relatively short decay period of existing studies and the potential impact of field aging, the long-term stability of biochar is still uncertain. Therefore, the influence of biochar degradation process on the remediation effect of biochar and the change of heavy metal adsorption properties of the biochars after degradation need to be further clarified. In addition, the long-term loading of biochar in the soil will leach a large amount of water-soluble organic matter, and then significantly change the contents and characteristics of dissolved organic matter in soil (Huang et al., 2019; Yang et al., 2019). Therefore, heavy metals adsorbed by biochar may also be released back into the soil with the leaching of water-soluble organic matter from biochar.

4.2. Potential abiotic factors

4.2.1. pH

The pH of soil is the key factor controlling the forms of the heavy metals in the soil (Dong et al., 2009). Biochar is usually alkaline, so after the application of biochar, pH of the soil will significantly increase, which is also one of the key factors for the biochar immobilization of heavy metals in the soil (He et al., 2019a,b; Dai et al., 2017). Rees et al. (2014) demonstrated that the heavy metals in the soil were immobilized by biochar within a short time mainly through the internal diffusion of the biochar particles and the increase of soil pH. However, the alkalinity of biochar and its long-term effectiveness in improving soil pH are questionable, for example, Jones et al. (2012) found that for the experiment of 3-year duration, there was a decrease of the pH of biochar, recovered from the soil, by 2 units. Lucchini et al. (2014) found that both the wood biochar and its ash content led to significant increase of the pH value of soil at the early stage of application. However, the liming effect was relatively short-term, with the pH of soil decreaseing to around the control value within about 7 weeks. Similar results were found in the study of de la Rosa et al. (2018), the pH of biochar decreased significantly with increasing aging time in the soil, reaching the pH value of the soil after 24 months (Fig. 2). In addition, have some studies also reported that the alkalinity of biochar decreases after water leaching over a period of time, with the long-term acid-rain leaching of biochar changing its nature from strong alkaline to acidic and thereby reducing its repair ability (Uchimiya et al., 2011). It suggests that the liming effect of biochar may be transient, so the effect of biocahr on heavy metals may also be transient. The field test results of Cui et al. (2016a) showed that after biochar treatment, the soil pH began to increase but is followed by a decline, but the change of bioavailable Cu and Cd concentrations was opposite, Its indicating the decrease of alkalinity of biochar caused by leaching reduced the biochar availability (Cui et al., 2016a). In general, the both of decrease in the alkalinity of biochar and the acidification of the soil environment may affect the long-term effectiveness of the immobilization effect of biochar on heavy metal contaminated soil.

4.2.2. Redox potential

In the actual field remediation process by biochar, the redox potential of the soil will be affected by various factors, especially flooding and drought conditions (Gonzalez-Mendez et al., 2017; El-Naggar et al., 2018; Sui et al., 2018). Moreover, the soil redox potential is closely related to the chemical forms of heavy metals in soil (Meng et al., 2019; Wang et al., 2019a,b,c; Antic-Mladenovic et al., 2017), because it directly or indirectly influence pH, DOC, redox chemistry of Fe, Mn, S,

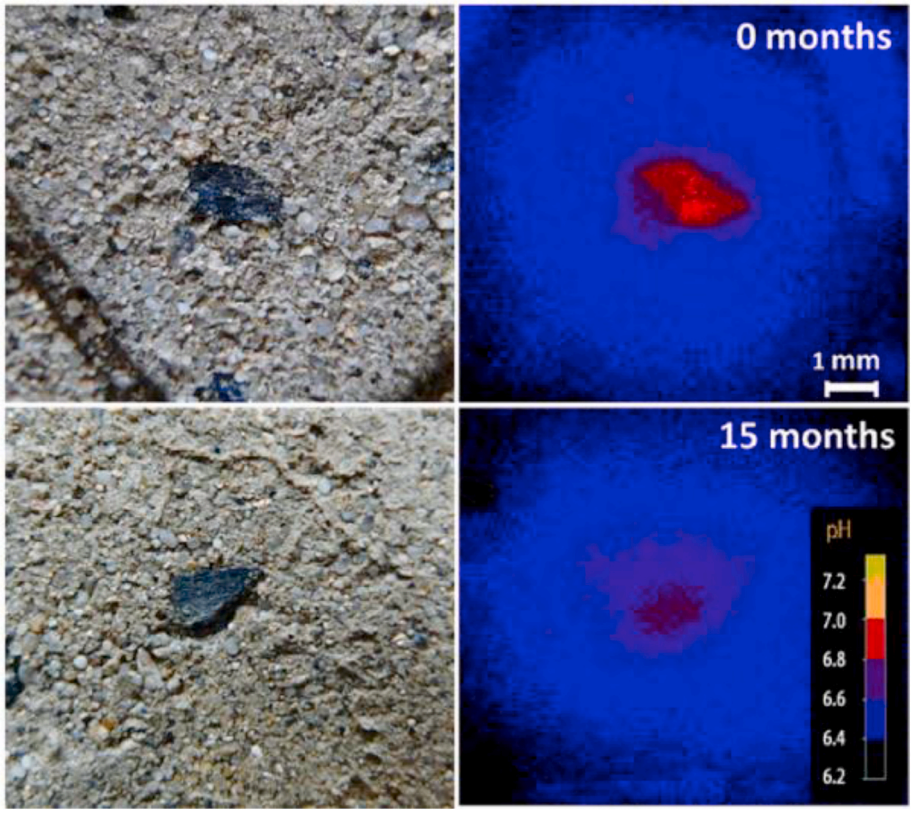


Fig. 2. The change of biochar pH after aging (15 months) in soil (Rechberger et al., 2019).

and N, microbial activities of soil (Schulz-Zunkel et al., 2015; El-Naggar et al., 2018).. Therefore, the effectiveness of the immobilization remediation of biochar may also effected by the soil redox potential. For example, El-Naggar et al. (2018) found that the contents of soluble Cd and Cu in the soil were significantly increased under aerobic conditions in comparison to the original value. Furthermore, the soluble heavy metal content in soil treated with biochar was significantly higher than that in soil treated without biochar (Fig. 3). The presence of biochar under aerobic conditions decrease the soil pH and increase the content of soluble aromatic carbon, which is one of the main reasons for the increase of soluble heavy metal content in the soil (El-Naggar et al., 2018). Therefore, when the soil redox potential changes, the immobilization effectiveness of biochar may be significantly reduced, and even the environmental risk of heavy metal release may be increased. Other studies have found similar results, the presence of rice husk biochar in the soil may promote the release of Mo, Co and As under flooded conditions (El-Naggar et al., 2019); In result of Beiyuan et al. (2020), the immobilization effect of pine sawdust biochar (300°Cof pyrolysis temperature) for Pb was seriously lacking in the environment having redox potential of −100 mv. However, the immobilization effect of pine sawdust biochar biochar (500°Cof pyrolysis temperature) for Pb did not change with the redox potential, which suggesting biochar produced at higher temperatures can be more suitable for immobilization of Pb under variable redox potential. According to previous research results, the main reasons why the redox potential affects the immobilization effect of biochar are: 1. Lower pH (biochar may increase the range of soil pH) (El-Naggar et al., 2018, 2019) 2. Dissolution of Fe/Mn oxides (Beiyuan et al., 2020); 3. Oxidation of sulfur (El-Naggar et al., 2018, 2019).

4.2.3. Cl^- concentration

It is well known that chloride ions can change the form and distribution of heavy metals in soil by forming metal-chloride complexes (Guo et al., 2018). Khoshgoftar et al. (2004) reported that Cl^- addition increased Cd concentration in the soil solution. It was also reported earlier that the presence of Cl^- would lead to the release of Cd from the soil (Bingham, 1984). Therefore, the presence of chloride ions may affect the immobilization effect of biochar. Compared to the washed biochar, Li et al. (2019a) found serious reduction of the immobilization effect of unwashed biochar during the process of remediation of Cd contaminated rice paddies. The possible reason is that the unwashed biochar releases a large amount of chloride ions in the soil environment, which decreased the soil pH, increased the DOC in pore water and promotes the formation of Cd^{2+} and Cl^- complexes, and then increased the bioavailability of Cd in the soil.. Li et al. (2019b) also believed that the biochar significantly increased the soluble Cl^- content in the soil environment. As a result, biochar treatment led to reduced content of available Cd in the soil but simultaneously led to significantly increased Cd accumulation in rice.

4.2.4. Other factors

In addition, acid rain may also affect the immobilization effect of biochar on heavy metals. Uchimiya et al. (2011) found that the Cu adsorption by biochar during soil remediation is a reversible process. There will be re-release of most of the adsorbed Cu under the presence of acid rain and acetate buffer and the irreversible adsorption capacity of Cu is dependent on the type of biochar. Wang et al. (2018) also found that acid rain can greatly inhibit the immobilization remediation of biochar. Therefore, for practical applications of biochar remediation, the risk assessments of heavy metal re-release needs to be fully considered. Various studies have indicated the positive correlation of DOM with the activation of some heavy metals (Beesley et al., 2010; Beesley and Dickinson, 2011). In the study of Yang et al. (2020), straw mulching produced a large amount of DOM, which inhibits the adsorption of biochar on Biophenol S under long-term action. Therefore, it is also necessary to study whether the generation of large amount of DOM

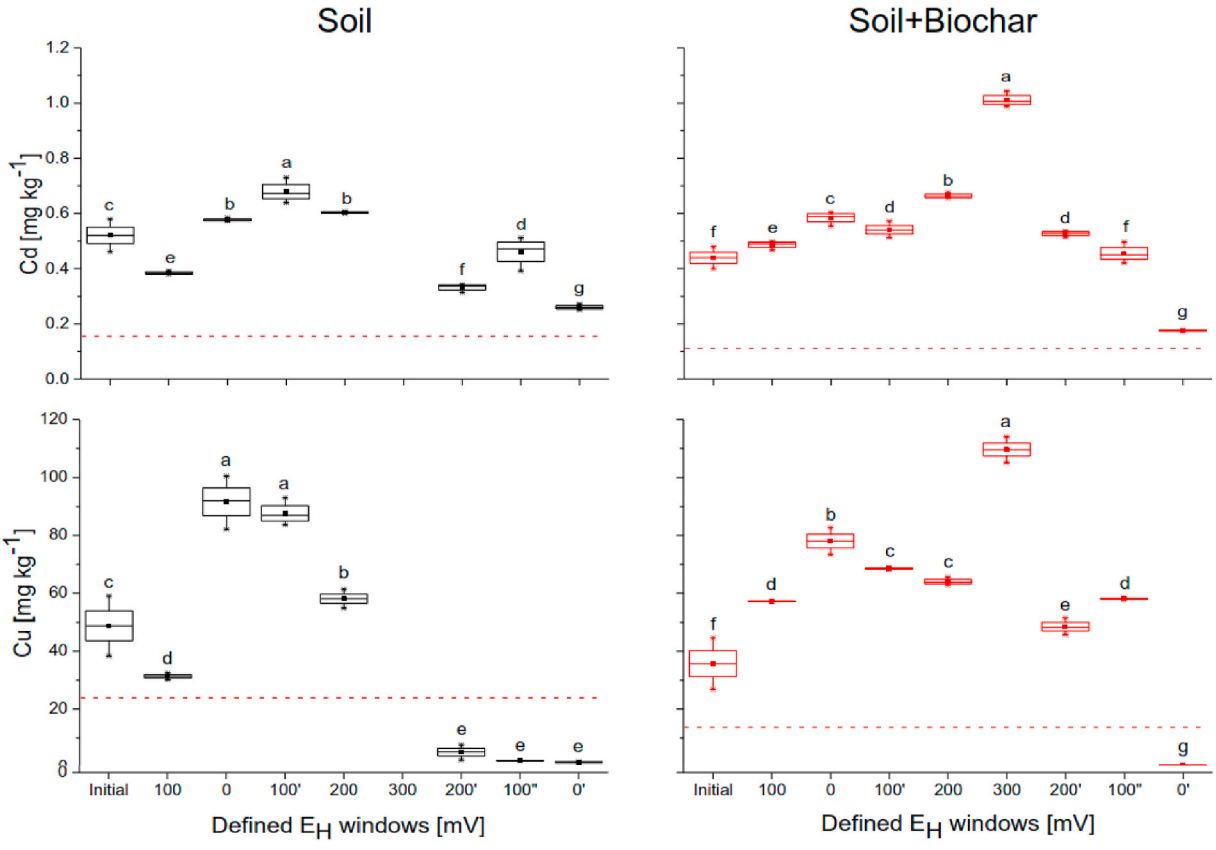


Fig. 3. Influence of dynamic redox conditions (Eh) on the phytoavailability of Cd and Cu in the non-treated soil and biochar-treated soil (El-Naggar et al., 2018).

during the straw mulching process will have any negative impact on the immobilization effect of biochar. For example, during a field experiment of two year duration, appilication of biochar reduced the Cd and Zn content in the soil solution by significantly increasing the pH of the soil. However, increasing content of DOM in the soil led to the increase of Cu and Pb content in the soil solution (Wagner et al., 2015). This also indicates that the large amount of DOM input from external sources may potentially reduce the immobilization remediation of biochar when repairing the multiple heavy metal contaminated farmland. Finally, it is a well known fact that the immobilization effect of biochar is often proportional to the amount of the applied biochar. Spokas (2010) observed breaking up of biochar when exposed to water and soil during a 24-h weathering experiment and simultaneous loss of the broken biochar with soil loss. Additionally, the biochar is blown away from the soil surface by the wind as a result of the low density of biochar, especially during dryland farming, with more dust getting generated from more quantity of the biochar (Chia et al., 2015 ; Li et al., 2018). The main reasons behind the loss of pyrolytic carbon in soil are erosion due to the environmental factors and the dissolution in the water flow with subsequent transportation (Jaffe et al., 2013). Therefore, the natural loss of biochar in the environment over a long period of time may also affect the restoration effect of the biochar.

4.3. Potential soil biotic factors

4.3.1. The influence of plant

During the application of biochar for soil remediation, it is inevitable to have the interaction between biochar and plant roots as the plant roots in the soil are dense. The rhizosphere is a micro-environment with unique physical and chemical properties, thereby having a certain influence on the properties of biochar and bioavailability of heavy metals. Ren et al. (2016) investigated the effect of wheat root system on the biochar aging process over a period of three months. It was found that the wheat root system significantly affects the physicochemical properties of biochar, and the degree of influence is dependent on the nature of biochar and its distance from the wheat roots. As we all know, root-induced changes in soil pH are recognized as a key mechanism influencing heavy metal bioavailability in the rhizosphere. Houben and Sonnet (2015) directly studied the immobilization effects of biochar in the rhizosphere environment. They found that *Agrostis capillaris* and *Lupinus albus* could decrease the immobilization effect of biochar on Cd, Pb and Zn in soil by reducing the pH of rhizosphere soil. Moreover, there was no reduction in the content of the heavy metals in the aboveground parts of the above mentioned plants due to the application of biochar (Houben and Sonnet, 2015). In addition, considerable number of studies have found that the root exudates can improve the bioavailability of the metals around the rhizosphere by promoting desorption of the residual metals (LeFevre et al., 2013; Zeng et al., 2017). For example, according to the study of Debela et al. (2010), presence of low molecular weight organic acids in the root exudates can significantly increase the activity of Pb in the contaminated soil. In addition, low molecular weight organic acids can promote the protonation of the biochar surface in the soil. Subsequently, the adsorption capacity of protonated biochar for the cationic heavy metals decreased, leading to re-release of a large number of cationic metals but the re-sorption of oxygen-containing anions of heavy metals (Alozie et al., 2018). Finally, the results of Oustriere et al. (2017) showed that the content of Cu^{2+} in the pore water of soil with biochar increased significantly after planting european black poplars for 22 months under the treatment of biochar. They speculated that the decrease of phosphorus in the pore water after planting and the decrease of the immobilization effect of biochar caused this phenomenon (Oustriere et al., 2017). In general, plant roots may affect the immobilization effect of biochar through accelerating the aging of biochar and changing the rhizosphere microenvironment.

4.3.2. The influence of earthworm

At present, earthworms are widely reported to significantly increase the bioavailability of heavy metals in the contaminated soil, which is contrary to the effect of biochar. Therefore, the presence of earthworms may affect the immobilization effect of biochar. For example, Gomezeyles et al. (2011) have found that the presence of earthworms significant increased the contents of water-soluble Co, Cu, Ni and Zn in the soil, and significantly reduced the immobilization effect of biochar. Wang et al., 2019a,b,c and Huang et al. (2020) also found that the presence of earthworms results in the significant increase of the available Cd, Pb and Zn (Cu) contents in the soil along with weakened immobilization effect of biochar. They concluded that the significant decrease of pH caused by earthworms was one of the reasons for the decrease of biochar immobilization effect. However, the results of some scholars have also showed that the activities of earthworms significantly increased the available heavy metals content in the soil while increasing the pH of soil (Wen et al., 2006; Udovic et al., 2007; Guo et al., 2020), suggesting that earthworms have different mechanism for activating the heavy metals. In previous research, some of the heavy metals adsorbed on the soil particles which have been eaten by the earthworms will be absorbed in their intestines and then secreted into the soil in different forms, resulting in the increase of the available heavy metal content in the soil (Vijver et al., 2003; Sizmur and Hodson, 2009). Moreover, there are considerable evidences to suggest that the earthworms can eat the biochar particles in the biochar mixed soils (Ponge et al., 2006). Therefore, will the heavy metals adsorbed by the biochar swallowed by earthworms have the same fate as the heavy metals adsorbed by soil particles? In addition, some scholars have also proposed that the increased accumulation of metals by plant may be a direct result of the formation of organometallic complexes by earthworms through the release of metal chelating organic substances (Wang et al., 2006; Wen et al., 2006; Udovic et al., 2007). Besides, Leveque et al. (2014) pointed out that the increasing soil porosity due to earthworm activities along with the leaching of deep soil by rainwater were some of the reasons behind the increase of bioavailable heavy metals. In the process of biochar remediation, earthworms eat biochar and soil mixtures containing absorbed heavy metals and transfer them to the surface soil and into the earthworm burrows. This also increases the risk of rainwater leaching, which may lead to reduced biochar immobilization effect (Uchimiya et al., 2011).

4.3.3. The influence of microorganism

The microorganisms are the most widely distributed and abundant soil organisms and most of them use ‘C’ as the energy source. Biochar is essentially carbonaceous substance and can be used by soil microorganisms in the soil environment as their energy source. Farrell et al. (2013) found that the soluble carbon part of biochar prepared from wheat and eucalyptus could be rapidly (3 days) consumed by the soil microorganisms. Hamer et al. (2004) found that during the 60-days microbial treatment experiment, there were loss of 0.8%, 0.7% and 0.3% C content from the black carbon obtained from corn, rye and oak, respectively, in the form of CO_2 emissions. However, there are few studies on the adsorption characteristics of biochar’s easily degradable or stable components on heavy metals. Therefore, it is necessary to further study the influence of microorganisms on the adsorption of heavy metals by the biochar during the process of biochar decomposition.

In addition, the microorganisms will directly or indirectly dissolve the heavy metals in the soil through their various metabolic activities. In the process of soil microbial metabolism, various low molecular organic acids will be secreted, and these metabolites will dissolve heavy metals (minerals) in the environment (Berthelin et al., 1995; Jing et al., 2007; Sessitsch et al., 2013). For example, *Bacillus subtilis* and *Bacillus pumilus* significantly increased the bioavailability of Cu and Cr concentration in the soil (Abou-Shanab et al., 2008). Some microorganisms can also release the heavy metals which bind to them via oxidizing anions, such

as thiobacillus ferrooxidans, which oxidizes the divalent sulfur anion in the soil, releasing some heavy metals as ions (Chaudhary et al., 2019). In addition, the rhizosphere microorganisms decrease the soil pH through nitrogen assimilation and subsequently increase the Cu^{2+} content in the rhizosphere soil (Cloutier-Hurteau et al., 2008). The above results indicate that microbial activity may increase the available heavy metal content in the soil and subsequently weaken the immobilization effect of the biochar. In addition, Xu et al. (2020) found that the release of Cr in rice rhizosphere was closely related to the recovery of Fe, and when the abundance of Geobacter, the main iron-reducing bacteria involved in Fe reduction, was significantly increased in presence of the pig manure biochar. In the meanwhile, quite a few studies have also showed that the biochar can increase the activity of the iron-reducing bacteria in the soil, and thereby promoting iron reduction (Qiao et al., 2018). For this reason, Xu et al. (2020) proposes that in the strategy of applying biochar to reduce the uptake of Cr in rice, the *Geobacter* and Fe reduction should be considered. Therefore, the presence of microorganisms may affect the remediation effect of biochar.

5. Conclusions and future perspectives

Currently, biochar is being widely used for in-situ remediation of heavy metal contaminated soil, especially in the agricultural land. However, most of the studies being carried out presently have focused on the laboratory conditions and short-term field remediation; with more or less all the studies have shown the effectiveness of biochar in reducing heavy metal migration and plant uptake from the soil. However, there are also some problems such as the negative effects of biochar on As-contaminated soil and the unreasonable of the large application amount of biochar in the field. Moreover, a certain number of studies have demonstrated the long-term effectiveness of biochar immobilization over a period of 2–3 years or even 5 years. However, some studies have found that the immobilization effect of biochar decreased with increasing aging time. The immobilization effect of biochar has problems such as instability and uncertainty. That soil environment significantly affect the actual immobilization effect of biochar, has also been reported by a large number of studies. For example, abiotic factors including soil pH, redox potential, DOM content, acid rain and biochar's physical and chemical properties, as well as biotic factors including earthworms, microorganisms and plant roots can significantly affect the actual immobilization effect of biochar. Summary, as a green and low-cost immobilization remediation material, biochar is very suitable for the remediation of heavy metal pollution of the agricultural soil. However, variable and complex environmental factors may seriously affect the immobilization effect of biochar. Therefore, a great deal of work needs to be done to find a more long-term and effective method for remediation of heavy metal contaminated soil by biochar. To this end, we highlight below some perspectives that related to the remediation of soil heavy metals by biochar for further study.

5.1. More long-term and regional field experiment

As displayed in Table .2, the field observation time in the field is basically concentrated for 2–3 years. Therefore, more long-term field experiments are required to study the long-term effectiveness of biochar. In addition, most of the studies were conducted on the rice fields. The uniformity of climate and soil type in the rice-producing areas and the uniformity of crops are huge limitations of biochar research. Thus, more regional soil remediation research is needed, especially in presence of extreme climatic conditions, such as the freeze-thaw cycles of seasonally frozen land, which may significantly affect the bioavailability of heavy metals in the soil and the effect of immobilizing reagents (Hou et al., 2020).

5.2. Improvement of biochar immobilization effect

As we all know, some biochar has the problems of limitations in the scope of remediation (for example, it is not suitable for the remediation of soil contaminated by As) and instability of the remediation effect. Therefor, many researchers have tried different approaches to solve these problems. For example, Wang et al., (2019a,b,c) found that iron-Modified magnetic biochar can simultaneously remove As, Cd and Pb from the contaminated soil. Zhao et al. (2020) through accelerated simulation experiment of long-term aging effects found that the sulfur-modified biochar had a more long-term stable immobilization effect on Hg, compared with non-modified biochar. In addition, in study of Beiyuan et al. (2020), the biochar produced at 550°C was more suitable amendment than it produced at 300°C for remediation of Pb-contaminated soils which were frequently flooded. Therefore, research on improving the immobilization capacity and long-term effectiveness of biochar through specific biochar preparation (different raw material, temperature and time, etc.) and modification technologies (acid treatment, base treatment and impregnation of mineral sorbents and magnetic modification) is of great significance.

5.3. The accelerated aging effect of soil organisms on biochar

When added to soil for remediation of heavy metals contaminated soil, biochar may react with soil constituents, such as soil minerals and organic matter, plant roots, geobiont and soil microbes (Yang et al., 2016; Ren et al., 2016; de la Rosa et al., 2018). For example, Ren et al. (2016) found that the physicochemical properties of biochar exposed to wheat roots for three months were significantly changed. The results of Quan et al. (2020) showed that Microorganisms promoted the release of soluble organic matter from biochar. However, in present, the research on the aging effects of plant roots and soil animals, such as earthworms, on biochar is very lacking. The effect of plant roots and earthworms on the physical and chemical properties of biochar and the immobilization effect of biochar for heavy metal is worth studying.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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